

The End Game in Kuhlthau's ISP Model: Knowledge Construction for Grade 8 Students Researching an Inquiry-Based History Project

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This article reports on a field study of the information behavior of Grade 8 students researching an inquiry-based class history project. Kuhlthau's 7-stage Information Search Process (ISP) model forms the conceptual framework for the study. The aim of the study was to define an end game for the ISP model by answering the following question: How do the student participants' feelings, thoughts, and information behavior lead to the construction of new knowledge? Study findings tentatively indicate that knowledge construction results from an iterative process between the student and information, which can be divided into 3 phases. In the first phase, the students formulate questions from their previous knowledge to start knowledge construction; in the second phase, newly found topic information causes students to ask questions; and in the third phase, the students answer the questions asked by this newly found topic information. Based on these results and Kuhlthau's own ISP stage 7 assessment definition of the ISP model end game, we propose a model of knowledge construction inserted as an extra row in the ISP model framework.

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Introduction

This article reports on the findings of a field study of the feelings, thoughts, and actions of 10 Grade 8 students as they seek, find, and use information for an inquiry-based class history project. The conceptual framework for the study is Kuhlthau's (1993b) Information Search Process (ISP) model, one of the most cited information-seeking models in information science over the last 20 years (Blessinger & Hrycaj, 2010; Levitt & Thelwall, 2009; Whitmire, 1999). The ISP model is a middle-range model (see Chatman, 1996, p. 19, note 1) devoted to describing and predicting the information-seeking behavior of one group of information users: students researching an inquiry-based school assignment.

The originality of the ISP model is that it divides the student's construction of the school assignment into six stages, thus linking the student's information need and consequent information seeking and use to the concept of temporality (Beheshti, Cole, Abuhimed, & Lamoureux, 2015). In this temporal approach, the student feels, thinks, and seeks information differently depending on the stage of his ISP. At its conception, the ISP model's temporal conceptualization of information need, seeking, and use was in contrast to the prevailing model of information need as a static state of mind (Bates, 1989).

The second innovation of the ISP model is that it describes the role of the student's feelings, thoughts, and actions in (a) constructing the school assignment and (b) directing the student's information-seeking behavior to effectuate this construction. There is a content side to information, but there is a procedural side to it as well—the perceptual and cognitive mechanisms that lead to the construction of new knowledge.

The aim of the study reported on in this article is to extend the Kuhlthau ISP model by investigating how a study group of Grade 8 students' feelings, thoughts, and information behavior combine to enable the students to construct knowledge on his school assignment topic. We divide the article into three main parts.

In the first part, we describe the conceptual framework for the study, Kuhlthau's seven-stage ISP model. The seven-stage ISP model adds insights into the assignment construction process, describing, through an examination of the feelings, thoughts, and information-seeking actions of the student, the end game of knowledge construction, which we define as the effect on the student of the knowledge constructed from information found while doing the assignment. In this part of the article, we also review two ISP studies to explore the issues surrounding how the students' feelings, thoughts, and information behavior combine together for knowledge construction.

In the second part of the article, we review the literature on the knowledge construction components of the ISP model, centering the review on Polanyi's (1966) theory of implicit-to-explicit knowledge conversion.

In the third and final part of the article, we report on the study of 10 Grade 8 students researching a history project over the 3-month period it took to complete the project. We propose a model of knowledge construction by inserting it as a new, fourth row inside the ISP model, which summarizes the study findings. We conclude the article with suggestions for further research that test the assumptions and hypotheses underlying this new fourth ISP row.

Conceptual Framework: Kuhlthau's Seven-Stage IPS Model

The conceptual framework for the reported study is Kuhlthau's ISP model. The better-known six-stage version describes the process of the student's knowledge construction over the course of researching a school assignment. The seven-stage version we use as our study's conceptual framework has an added "assessment" stage, which takes place after the student has handed in the finished school assignment. It asks the students about their own metacognition: How do students feel, and what do the students think as a result of the newly constructed knowledge they have learned from doing the school assignment?

The ISP model conceptualizes information seeking inside a constructivist learning theory framework. Starting from the student's own previous knowledge of the assignment topic, which includes implicit knowledge—automatic

human procedures for acquiring new knowledge—students then seek, find, and use information to construct their assignment.

Constructivist learning theory, in turn, is based on constructivist theory, which, generally speaking, attempts to explain how humans come to know what they know. According to this broader theory, individuals construct knowledge through their interactions with the environment. Knowledge is actively constructed by the individual, who is seen as being inside a specific time and space (Wilson, 1997). Inside this time and space, the individual is motivated to make sense of his world, to develop meaning and an understanding of this world (Novak, 2011), the process of which, because it is personal, can be described as developing the person's "perspective" on the world (Murphy, 1997, p. 4). This is an important, fundamental Kuhlthau principle, that focus formulation is the student identifying from "the information gathered" his personal perspective on the topic (Kuhlthau, 1983, p. 4).

Applied to the education environment, the constructivist perspective shifts school learning from memorizing facts from an authoritative textbook to the individual student investigating topic information from diverse sources for the purpose of constructing his personal perspective on the assignment topic. (For a checklist for constructivist learning theory applicable to the education environment, see Murphy [1997]). In constructivist learning theory, the coming together of the student's previous knowledge and new information found in the environment—in books, articles, and websites—has been primarily investigated at a theoretical level by Ausubel (1962, 2010; Ausubel & Ausubel, 1969; Ausubel, Novak, & Hanesian, 1978).

Kuhlthau's particular constructivist theory influences have been Bruner (1974), Dewey (1997), Kelly (1963), Vygotsky (1986), and, through Bruner, Jean Piaget (1972). Although the cognitive development concepts and stages espoused by Piaget have been subsequently criticized (Parent, Normandeau, & Larivee, 2000), they still form the basis for discussion and research on this topic. We briefly describe Piaget's theory here.

According to Piaget (1972), humans are in a constant exchange with the environment in order to create a balance through constant adaptation to the environment to achieve equilibration (Huitt & Hummel, 2003). Humans effectuate equilibration through knowledge or cognitive structures called schemata (schemas) and through two processes Piaget calls assimilation and accommodation (Lutz & Huitt, 2004). Assimilation is humans modifying the environment so that it will fit their pre-existing knowledge structures (see the frame theory of Minsky [1975]); accommodation is the process by which individuals modify their knowledge structures to adapt to unfamiliar environmental input. In learning, individuals must detach from the tried and true knowledge structure and assimilation and shift to the modification of their knowledge structure and accommodation.

The center of Piaget's theory of child development is that children transition to adolescence and then adulthood by

making accommodations to unfamiliar environmental input in four stages of cognitive development: (a) sensorimotor (infancy); (b) preoperational (early childhood); (c) concrete operational (elementary and early adolescence); and, finally, (d) formal operational (adolescence and adulthood). According to Siegler (1991), those individuals who reach the formal operational stage can think abstractly about truth, the nature of existence, and “provide alternative, competing beliefs about these” (Lutz & Huitt, 2004, p. 72; see also Steinberg, 2005; Steinberg et al., 2009).

The transition between concrete and formal operational stages is of primary importance for the present study’s 14-year-old student participants whose class assignment required them to come up with a thesis focused on a self-selected history topic. According to one study (Eylon & Linn, 1988), only 23% of 14 year olds in the study reached the formal operational stage, a percentage that only increased to 34% by the end of high school (see also Kuhn, Langer, Kohlberg, & Haan, 1977), which contradicts Piaget’s contention that most, if not all, adolescents reach the formal operational stage by age 15 (Lutz & Huitt, 2004). This opposing contention is that, contrary to Piaget’s belief that biological development pushes all children by age 15 into the formal operational stage of development through normal interaction with the environment (Jordan & Brownlee, 1981), the transition from concrete to formal operational is actually dependent on the child’s environment (Lutz & Huitt, 2004; see also Naidoo & Naidoo, 2007). We return to this last research finding in the Limitations section of this article.

Kuhlthau’s (1991) six-stage ISP model sets out a course for knowledge construction through the interaction between the student and information. The students feel uncertain and think vague thoughts in ISP stage 1, when the class instructor initiates the students’ class assignment by announcing

assignment requirements. The students’ thoughts become clearer and they feel less uncertain in stage 2 when they select their assignment topic. Uncertainty increases in stage 3 as the students explore information on the topic. Contrary to the “axiom” that information reduces uncertainty, at this stage, the information the students find actually increases their uncertainty (Kuhlthau, 2004, p. 200). The students’ uncertainty decreases in stage 4 when they formulate their personal perspective or focus on the topic. Their thoughts become clearer, which affects positively the efficacy of their information seeking. They also become more interested in their topic and their assignment. Thereafter, in stages 5 and 6, as they collect evidentiary information in support of their focus, the students’ feelings of uncertainty diminish and their confidence increases. Kuhlthau’s (1991) six-stage ISP model is a description of the stages of the students’ knowledge construction for a class assignment.

Kuhlthau applied constructivist theory to practice in her seven-stage ISP model, which adds a learning assessment stage to the better-known, six-stage version. The seventh-stage assessment asks students to look back and consider the process of knowledge construction over the course of doing a class assignment, which enables the student to then transfer this metacognition of their “higher order thinking” to the next case (Kuhlthau, 2004, p. 4). Although student self-assessment was present as part of stage 6 as early as Kuhlthau (1985), the seven-stage version of the ISP model and the separate assessment function it encapsulates is clearly expressed in the second edition of the 1985 book (Kuhlthau, 1994, p. IV), and expressed in chart form as shown in Figure 1 (following) in Kuhlthau, Maniotes, and Caspari (2007, p. 19). For an example of the assessment stage in the field, see Harada (2005).

In this article, we highlight students’ assessment of their learning and the seven-stage version of the ISP model,

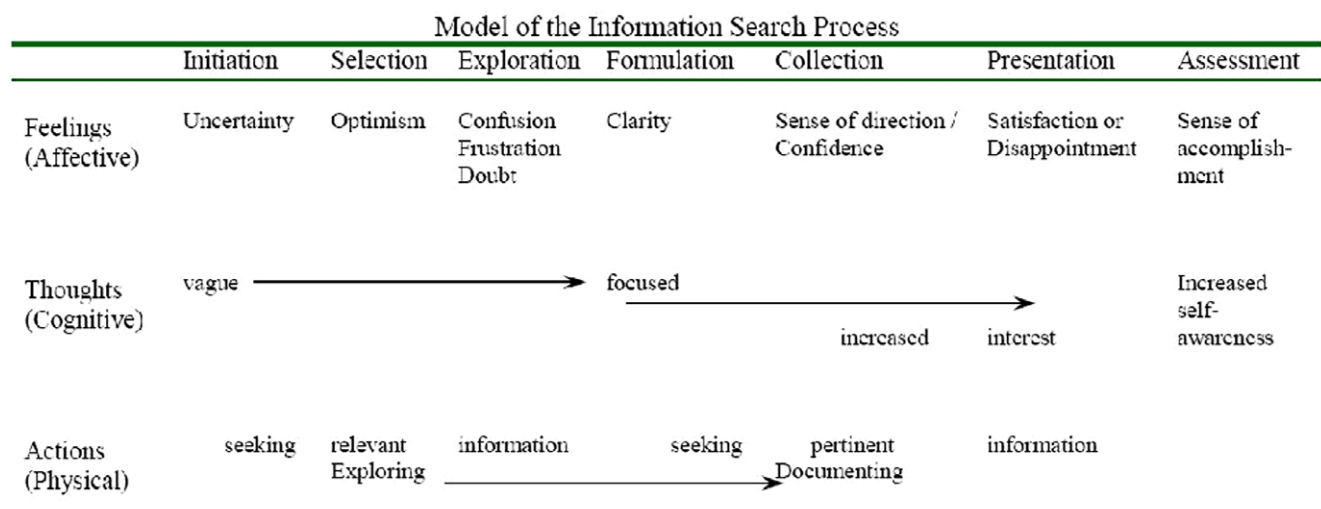


FIG. 1. Kuhlthau’s seven-stage ISP model (from Kuhlthau et al., 2007, p. 19). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

shown in Figure 1, because it gives essential insight into Kuhlthau's end game conceptualization of how and why students seek, find, and use new information to construct new knowledge for their school assignment. While the six-stage ISP model is predominantly descriptive, in the seven-stage version there is an element of cause and effect; however, the cause occurs only if there is a successful constructive learning process resulting from the assignment, with the seventh stage indicating the effect of successful knowledge construction. As shown in Figure 1, the seventh stage result of the six-stage process and the presentation of the finished assignment is a "sense of accomplishment" and "increased self-awareness."

Components for Construction

Uncertainty and the Student's Feelings, Thoughts, and Actions

A basic touchstone in information science is that information seeking is motivated by a gap or anomaly in the individual's knowledge (e.g., the ASK concept in Belkin, Oddy, & Brooks [1982]), which causes an increase in uncertainty and anxiety. The individual then seeks and finds information that diminishes the uncertainty. From this comes the popularly cited equation: Information equals the reduction of uncertainty (e.g., Wilson, Ford, Ellis, Foster, & Spink, 2002; see also Marchionini, 2010). Kuhlthau (1993a, 1993b) found, however, that far from reducing uncertainty, the student's exploration of information on their topic in stage 3 of their ISP actually increases the student's uncertainty. Only in stage 4, with the student's formulation of the assignment's focus, does uncertainty decrease (then continues to decrease through stages 5 and 6 of their ISP).

Does information increase or decrease uncertainty? Wilson et al.'s (2002, p. 708) findings point in two directions: They found that their study participants' uncertainty (a) decreased steadily through the four problem stages, but (b) increased during a single search session with an information system (see also Moshfeghi & Jose, 2013). Wilson et al. (2002) concluded that, because of the cognitive-affective components of the uncertainty concept, it is problematic to require study participants to rate their uncertainty. Do they feel more uncertain or do they somehow observe in themselves that, as a result of the new information they have found, the state of their thinking is more uncertain? Or is it both? (Thellefsen, Thellefsen, & Sørensen, 2013, p. 1736; see also Marchionini, 2010).

Wilson et al. (2002) attempted to address the conundrum of the affective-cognitive aspects of uncertainty by separating out cognitive and affective uncertainty. They observed that many of the feelings in Kuhlthau's ISP model—listed in the first column of Table 1—are, in fact, synonymous or semisynonymous terms for feelings-uncertainty, which they put in the following continuums: pessimistic/optimistic; confused/clear; frustrated/relieved; doubtful/confident; dissatisfied/satisfied; and disappointed/pleased (Wilson

TABLE 1. Kuhlthau's seven-stage ISP model feelings, thoughts, and actions (from Kuhlthau et al., 2007, p. 19).

Feelings	Thoughts	Actions
Uncertainty	Vague	Seek relevant information
Optimism	Focused	Explore
Confusion	Increased interest	Seek pertinent information
Frustration	Increased self-awareness	Documenting
Doubt		
Clarity		
Sense of direction		
Confidence		
Satisfaction		
Disappointment		
Sense of accomplishment		

et al., 2002, p. 710). Wilson et al. asked their study participants to score their feelings using these continuums. Wilson et al. then correlated the study participants' separate uncertainty scores with these feeling scores, per problem stage. The highly significant correlations (i.e., at the .01 level) were:

1. *Problem recognition stage*: Uncertainty is highly correlated with disappointed/pleased;
2. *Problem definition stage*: Uncertainty is highly correlated with confusion/clarity and doubtful/confident;
3. *Problem resolution stage*: Uncertainty is highly correlated with pessimism/optimism and doubtful/confident; and
4. *Solution statement stage*: Uncertainty is highly correlated with confusion/clarity, doubtful/confident, dissatisfied/satisfied, and disappointed/pleased (Wilson et al., 2002, pp. 711–712).

Oscillation toward cognitive uncertainty. From the correlations just given to problem stage, Wilson et al. concluded that, among the semisynonymous words Kuhlthau associated with affective uncertainty, "doubt" and "confusion" should instead probably be affiliated with cognitive uncertainty (Wilson et al., 2002, p. 712). This is an interesting question: Does a study participant who is asked to rate his uncertainty rate his feeling of doubt and confusion or is he observing his thinking on a topic, issue, or problem? It is possible that the study participant oscillates between cognitive and affective uncertainty depending on which ISP stage he is when the researcher asks.

Let's take an example from the listing of Wilson et al.'s (2002) problem stages and the synonymous and semisynonymous terms highly correlated with uncertainty. If we assume to be true Wilson et al.'s observation that doubt and confusion are probably associated with cognitive, rather than affective, uncertainty, then their study participants in the second stage, the problem definition stage, were oscillating toward cognitive uncertainty and were

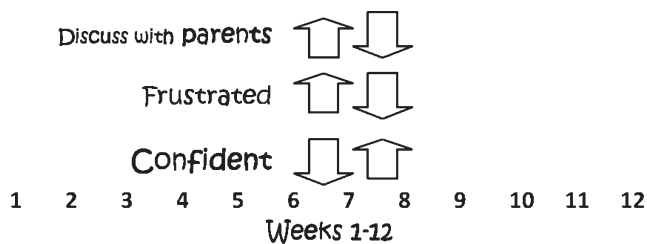


FIG. 2. Variance in the action “discuss with parents” and the two statistically significant feelings variables (week 6 to 7 compared to week 7 to 8), from Beheshti et al. (2015).

therefore in a predominantly cognitive stage of problem solving.

Oscillation toward affective uncertainty. It has been observed in previous ISP studies that, after undergoing a decline in their uncertainty from stage 1 to 2, student participants undergo an upsurge in uncertainty in stage 3 of their ISP when they are predicted to be exploring information on their topic (e.g., Kuhlthau, Heinström, & Todd, 2008).

In a recent field study of Grade 8 students researching a history assignment, Beheshti et al. measured the original Kuhlthau feelings, thoughts, and actions variables listed in Table 1. Additional action variables were also measured, including:

- _____ Discussing my ideas with my classmates
- _____ Discussing the assignment with my parents/brothers/sisters
- _____ Discussing my ideas with my teacher

To measure the temporality or evolution of these additional action variables, as well as the Kuhlthau variables for feelings and thoughts listed in Table 1, 12 separate measurements were taken, one measurement for each week of the 12-week duration of the study participants’ history assignment. Figure 2 summarizes one part of that study’s findings only: the up or down movement of the study participants’ feelings, thoughts, and actions in roughly the middle of the students’ 12-week process of researching and writing their assignment, from week 6 to 7, and again from week 7 to 8. In Figure 2, we highlight only the statistically significant variances from those weeks.

In Figure 2, the participants’ confidence decreased in week 6 to 7 and then increased in week 7 to 8. It is noteworthy that the variance in confidence of the student participants was not statistically significant for any of the 12 weeks, except the increase in confidence shown in Figure 2 from week 7 to 8. For other variables, the only statistically significant variances were:

- The decrease in student participants’ frustration from week 7 to 8 and
- The decrease in discussion with parents from week 7 to 8.

Beheshti et al. tentatively concluded that, in week 7 to 8, the student participants had overcome their crisis of confidence from the previous week.

During the students’ week 6 to 7 crisis in confidence, their increased need to discuss their assignment with their parents may signify that the student participants’ uncertainty oscillated toward affective uncertainty during week 6 to 7. We say this because, among all the possible channels or sources of information, parents are a very particular kind of channel of information seeking (Spink & Cole, 2001). They are a “sounding board” against which the students can bounce back ideas about their assignment (Newman, 2000), but a sounding board with a large emotional component to the feedback they offer their child.

As shown by “evidence of concept” from Wilson et al. (2002) and Beheshti et al. (2015), the possibility that participant uncertainty oscillates between cognitive and affective uncertainty, depending on when in the ISP the participant is, has potential implications for facilitating knowledge construction by effective information system design in particular and the provision of information services in general.

Literature Review

Knowledge construction process

The aim of this literature review is to establish a constructivist basis for the study that follows. In constructivist learning theory, knowledge construction is a process of new knowledge acquisition starting from the student’s previous knowledge. Previous knowledge gives initial direction to the student’s information seeking and processes new information for the goal of constructing new knowledge. Information and the automatic knowledge construction procedures the student knows but does not know he knows when he applies them—defined as implicit knowledge—are key ingredients in this construction process.

Implicit knowledge is procedural knowledge (Reber, 1992a, 1992b; Sherry & Schacter, 1987; Squire, 1992); in other words, the rules, functions, mechanisms, and procedures of *how* we know, including the unconscious and automatic “application of a procedure” for the performance of a task (Dienes & Perner, 1999, p. 744). Procedural knowledge includes a categorization process that requires anterior steps:

1. Priming procedures, which, in turn include:
2. Subception,
3. The “shock”
4. Categorization procedures (Knowlton & Squire, 1993)—that is, how we naturally “abstract information across encounters” to form a category in memory (Knowlton & Squire, 1993, p. 1749; see also Medin & Rips, 2005).

We make the assumption that priming-to-categorization is a process governed by implicit or unconscious knowledge that the individual is not aware of knowing and automatically and unconsciously applies to the processing of unfamiliar environmental data (including information in text).

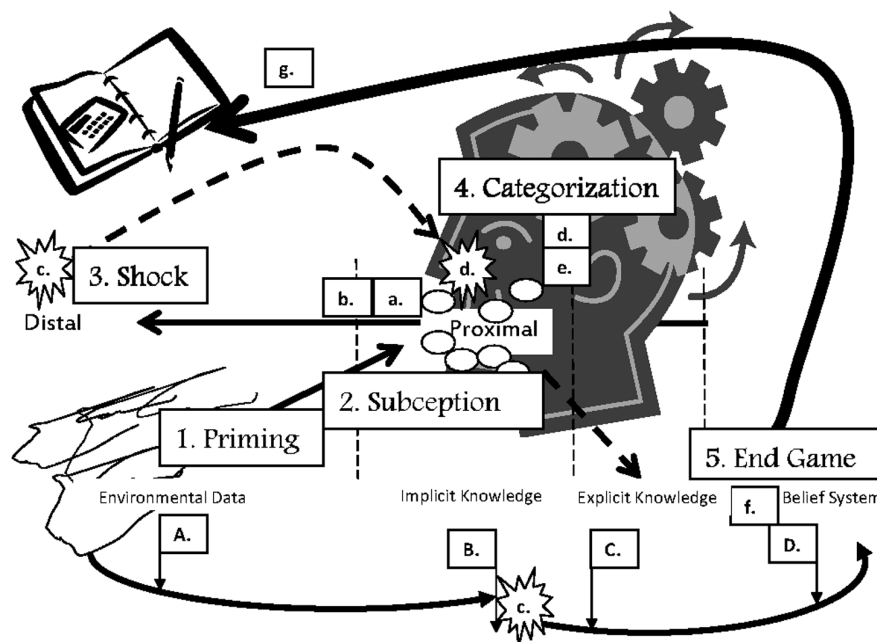


FIG. 3. Knowledge construction process whereby the student constructs new explicit knowledge from environmental data through implicit knowledge processes.

The priming-to-categorization process, in turn, is part of a larger process, the end game of which is new knowledge construction that the student can consciously utilize in a school assignment. We divide the end game into these two subsidiary parts:

5. End game: as a result of a knowledge construction process, the student's:
 - Belief system becomes actualized, added to and/or modified;
 - Feelings and thoughts flow together.

In the following paragraphs that describe these five stages of the knowledge construction process, the capitals, small letters, and numbers correspond to the capitals, small letters, and numbers of the diagram of the construction process in Figure 3.

1. *Priming* is a type of implicit memory procedure where facts or words that are unconsciously stored in memory are unconsciously recalled to affect subsequent decision making (Knowlton & Squire, 1993). In ISP stage 3, the student explores information on their newly selected assignment topic. This newly found topic information is "anchored" in ideas and concepts of the student's pre-existing cognitive structure (Ausubel, 1968). These anchors are called "subsumers" (de Castro, 2013). The subsumers recognize new information the student finds and then bind it to the student's cognitive structure. Because all subsequently found information must match the subsumers, they determine the external information that can be received (Novak, 2010). Over time, interacting with newly found information, however, the student's

receptor subsumers become increasingly modified, thus ready to receive increasingly differentiated incoming environmental information, in a process Ausubel (1968, p. 79) calls "progressive differentiation."

2. *Subception*: At a certain point of differentiation, subsumers become "particulars" of the student's assignment topic, enabling "subception" (Polanyi, 1966, p. 7). Subception is the "apprehension" of a relation between a group of particulars and subsequent information the student finds in a book or on the Internet. Subception is (a) particulars of previously acquired information anchored to the student's subsumers, called the proximal term, coming into a relation with (b) subsequently found information from the outside environment (in texts or in conversations with teachers, classmates, or parents), called the distal term:
 - a. The proximal term is acquired by the student through priming procedures and dispersed in the student's memory as disparate elements or "particulars," anchored in the subsumers. At this point, these particulars cannot be comprehended by the individual as parts of a whole entity.
 - b. The proximal term, however, unconsciously attunes the individual to the oncoming distal term, "signifying the approach" and readying the approach of the distal term (Polanyi, 1966, p. 11).
3. The "shock": Polanyi (1966) adopted the terms proximal and distal from the "shock syllable" experiments conducted in the 1940s. In these experiments, subjects were shown nonsense syllables that, unbeknownst to the subjects, heralded a physical shock administered to the subject by the researchers. These nonsense syllables came to alert the subjects to shocks that were about to be administered by the researcher, thus their label shock syllables.

Although the subjects showed signs of anticipating the shocks when the shock syllables were shown, they could not identify the syllables when asked to do so because they were paying attention to the shock itself. Because the experiment subjects attended to the distal term (the electric shock), they “disattended” to the proximal term (the shock syllables).

c. The distal term, the shock, occurs.

4. Categorization: According to Polanyi, the proximal term actually has a sort of explanatory function contained within its structure, which is dispersed or divided up into its particulars until the distal term shocks the disparate particulars into coherence (from the individual’s perspective). This occurs in two steps:

- d. First, “an intimation of the coherence of the hitherto not comprehended” disparate particulars/elements of the proximal term (Polanyi, 1966, p. 21). These intimations of coherence also produce:

- Intimations of meaning (thoughts-uncertainty) (Polanyi, 1966, p. 13), which leads to:
- A hypothesis set indicating alternative meaning interpretations. The number of hypotheses in the set is, at first, large, creating a feeling of unease-uncertainty (Thellefsen et al., 2013, p. 1740).

- e. Second, “an integration” of the heretofore disparate particulars/elements of the proximal term into “a coherent entity to which we are attending” (Polanyi, 1966, p. 18; see also Prosch, 1986, p. 70).

Categorization, essentially, is this quality of the disparate particulars anchored in the students’ previous knowledge subsumers coming together into a coherent entity, which the individual can now recognize as meaningful.

5. End game: The key to the end game of knowledge construction in the ISP model is Kuhlthau’s uncertainty principle. According to Kuhlthau’s uncertainty principle, uncertainty has both a feeling and thought component, with thought-uncertainty causing feeling-uncertainty (Kuhlthau, 2004, p. 200). The stage 3 rise in the feeling of uncertainty is a positive and necessary sign that the student is cognitively readying himself to become engaged with the topic during focus formulation in stage 4 (Kuhlthau, 1999; see also Wilson et al., 2002, p. 705).

The stage 3 rise in uncertainty is an unpleasant process, which constitutes a difficulty, even a barrier, to learning (Harada, 2002; Isen, Daubman, & Gorgoglione, 1987; Kahneman, 2011, pp. 59–60, 69; Kracker, 2002; Kwon, 2008). But, it is a process with high rewards: The new information sought and found as a result of the stage 3 increase in uncertainty, by revealing to the student his personal perspective on the topic, also enables the individual to “understand” part of his own belief system (Yan & Davison, 2013, p. 1145), which Kuhlthau (2004) calls, following Kelly (1963), the student’s personal construct system.

The end game in Kuhlthau’s ISP model is defined in the students’ stage 7 assessment of their feelings and thoughts after they have handed in their finished assignment. The

student feels and thinks, respectively, a “sense of accomplishment” and an “increased self-awareness” (Kuhlthau, 2004). (Guo & Klein [2009] describe the end game similarly in terms of six constructs: concentration; perceived control; merging of activity and awareness; transformation of time; transcendence of self; and autotelic experience.) With the criterion of “self-awareness,” Kuhlthau’s ISP model inscribes the student’s interaction with information to construct a thesis-based school assignment in the wider human quest of inductively constructing knowledge from experience in the world.

An influential definition of knowledge is Gettier’s (1963) “justified true belief” (Foley, 2002), which echoes the definition of knowledge since Plato (Faucher, Everett, & Lawson, 2008b). According to this definition, we believe something to be true, and we justify the belief through reasons in support of the belief gained through inductive experience in the outside world (Popper, 1979). However, this rather objectivist definition of knowledge (Kimble, 2013) is contrary to the constructivist framework we have taken in this article, particularly our focus on implicit knowledge, which is knowledge unjustified in reason, as an integral component of knowledge construction. Generally, there has, in fact, been a search for a broader, less intellect-based definition of knowledge (Foley, 2002, p. 178).

To this end, we define the end game of the student’s knowledge construction process as the actualization of self-knowledge of their belief system about the world, but particularly their assignment topic (i.e., rendering it explicit so that is consciously available to the student) by adding to and/or modifying their belief system. The student should consciously have access to their new knowledge for use in their school assignment. And, we highlight the role played by the student’s emotions or feelings, which are implicitly and unconsciously involved in knowledge construction, both in the process and its final form.

The shifting roles of emotion, information, and previous knowledge in the creation of new knowledge or meaning is given by Thellefsen et al. (2013, pp. 1739–1740) in a conceptual example of a man identifying a dot on the horizon coming toward him in three stages. In the first stage, when the object on the horizon is nothing but a spot, “emotion,” defined as the feeling of unease rather than in terms of uncertainty, is the dominant component of the man’s meaning-creation. In the absence of information and knowledge about what the dot is, the man creates a hypothesis set to explain the dot. It is this possibility of alternative hypotheses in the set that creates the man’s feeling of unease. The emotion drives the man to seek information from the spot, clues to its identity.

In the second stage, when the man searches for and finds clues to identify the spot as an unspecified dog, “information” is the dominant component. And, finally, in the third stage, when the spot is recognized as the neighbor’s nice and friendly dog named Jake, the man’s “knowledge system” is the dominant component in the meaning-creation process. However, there is still an emotion component attached to

the man's meaning-creation process even in this final, knowledge-dominant third stage because "the feeling [of unease] . . . evolve[s] into a state of joy when we explicitly come to think about Jake" (Thellefsen et al., 2013, p. 1741). Yan and Davison (2013, p. 1146) state that there is an increase in "flow," which they define as the person operating "at full capacity" when they are in a "dynamic equilibrium" (Nakamura & Csikszentmihalyi, 2002, p. 90) (see also Feist, 2006; Guo & Klein, 2009; Hoffman & Novak, 2009; Koufaris, 2002). Burton (2008, p. 4) calls this last stage when emotion and thinking flow together as "the feelings of knowing," which he distinguishes from mood feelings because "rather they feel like thoughts." In Figure 3, the end game is shown as:

- f. Actualization, adding to and/or the modification of the student's belief system, leading to the free and fluid flow between feelings and thoughts. In this way, the student achieves Kuhlthau's assessment stage 7 "feeling of accomplishment" and "increased self-awareness."
- g. Utilization by the individual of the new, explicit knowledge, of which the individual is now conscious and aware, enabling the student to communicate the new knowledge in a school assignment.

Indicated in the lower part of Figure 3 as a curved line broken into two parts, the knowledge construction process starts as (a) environmental data, undergoes (b) implicit knowledge processes, which turn data into (c) explicit knowledge, the end game of which is integration into the individual's (d) belief system. Following Polanyi (1966; see Kimble, 2013), we break the curved line after item b (implicit knowledge) to indicate that it is not a continuum. (For the continuum view, see particularly, Nonaka and Takeuchi [1995]; but also, Alavi & Leidner [1999] Faucher, Everett & Lawson [2008a] Monroe & Lee, [1999]; Saint-Onge, [1996]).

The Study's Specific Problem and Research Questions

The specific problem we set out to investigate in the study is the role of the feelings, thoughts, and information behavior of the study participants as they construct, then use, their new knowledge to create a history project. In effect, the explanation of how and why students construct explicit knowledge is the end game of knowledge construction in Kuhlthau's ISP model, which is not currently explicit in the ISP model. The three research questions underlying the study's data analysis are:

1. Do the data support Kuhlthau's ISP model stages?
2. Do the data support the evolution of the Kuhlthau ISP model's feelings, thoughts, and actions?
3. How can we chart the study participants' implicit-to-explicit knowledge construction, including the vital preparatory priming function, as it evolves over ISP stages?

The Study

The study reported here is the first phase of a 4-year research project whose ultimate aim is to design an information interaction intervention agent for Grade 8 students researching an inquiry-based history project for a class assignment. In this first phase, the researchers wished to conduct a field study of the information behavior of Grade 8 students from the beginning to end of their research process for a history assignment. Over several years before the study began, the researchers had had an association with a Canadian school with a well-developed Grade 8 history assignment, which would provide unlimited access to us. We define "well-developed Grade 8 history assignment" to mean that the assignment was already in place and completely independent of the researchers' input, thus providing a naturalistic research setting for the study. Several years before the start of our investigation, the school had developed an inquiry-based history project for all Grade 8 students at the school. The students were required to select a history topic and then, through interaction with information, develop a thesis. The topic-to-thesis structure is a mechanism of teaching a "constructivist perspective" (Hernández-Ramos & De La Paz, 2009, p. 167; see also De La Paz, 2005). The school's history assignment therefore agreed with the constructivist learning theory underlying the conceptual framework of our research.

In 2010, from a total of five Grade 8 history classes at the school, two classes were chosen by the teachers to participate in the assignment. The two classes had separate, but collaborating, instructors. The school then contacted the students' parents to inform them about our study. Consent forms were distributed among students to be signed by both their parents/guardians and themselves. The consent forms included a statement stipulating that the children would be able to participate in the research study only if both the parents/guardians and the student gave written consents. It also emphasized that, at any time during the research study, students could withdraw for whatever reason and the anonymity of students and the school would be maintained (i.e., only the research team would access the collected data), and any publications or public presentations of the research would make no direct reference to either the school or individual students.

In February–May 2011, we collected data using multiple data collection instruments (with different research questions for each instrument) from a total study group of 44 Grade 8 students (41% girls and 59% boys). Study findings from these other data sources have been previously reported in Beheshti et al. (2015) and Cole et al. (2013). The 10 volunteer study participants for the study (5 boys and 5 girls) we report on in this article were a subset of the larger total study group of 44 Grade 8 students.

The McGill researchers initiated the study of the 10 students in week 2 of their history course, which was the first week of March 2011. Over the next 3 months, the 10 students were interviewed at six different points; the last

TABLE 2. Progression over six interviews of Kuhlthau's seven-stage IPS model: first-order coding.

First-order ISP stage Interviews (1–6)	Initiation	Selection	Exploration	Formulation	Collection	Presentation	Assessment
1. March 1/3, 2011 Week 2 Spring break—2 weeks	1	119	0	0	0 Info.Seek 58	0	0
2. March 29/31, 2011 Week 4	0	6	40	44	7 Info.Seek 35	0	0
3. April 14, 2011 Week 6	0	0	14	30	0 Info.Seek 21	0	0
4. April 20/28, 2011 Week 7/8	1	0	1	48	4 Info.Seek 15	9	0
5. May 4, 2011 Week 9 History presentation, May 26	0	1	0	14	5 Info.Seek 8	29	0
6. May 27, 2011 Week 12	0	0	1	0	0 Info.Seek 71	0	128

interview was on May 27, 2011, which was the day after the students had presented the final version of their assignment. The students had a 2-week spring break after our first interview, which we subtracted from the study. This meant that the total duration of the students' history assignment and our research was 12 weeks. The chronology of the interview dates is shown in Table 2.

The final history project presentations took a variety of forms: a documentary video; an historical newscast; the performance of a skit; an art exhibit; an historical re-enactment; or any other agreed-upon presentation format. Most presentations included a PowerPoint presentation outlining the students' topic and thesis. The presentation constituted 40% of the history assignment mark. The teachers emphasized that the research process for the history assignment was more important than the final product—the instructors told the students the project was about “learning to learn”—and assigned 60% of the total mark for the assignment to hand-ins for (a) the students' history assignment topic and (b) a written proposal that contained the students' thesis statement and the structure and historical information supporting the thesis.

The study took place in the field in a real-life classroom setting and was not an experiment. The McGill researchers had no influence over either the course requirements or what the instructors said to their students. The students were not restricted to a topic list by their instructors, although the instructors extensively discussed topics with the students during class. The instructors encouraged the students to select a topic they would enjoy and would find interesting. The students were encouraged by the instructors to think outside their box (i.e., for the example topic the mafia the students were given the thesis statement: The Mafia is a necessary evil). This conforms to the constructivist learning principle that new learning takes place on the outer edge of what the student already knows, and that new knowledge construction starts from the previous knowledge, interests, and experiences of the student. Task completion and personal growth

are conflated, according to both constructivist theory and the practice of the instructors in the present study.

Data Collection

The aim of the present study was to obtain data that would describe, and perhaps explain, the metacognition of the 14-year-old study participants concerning their feelings, thoughts, and actions as they evolved over the 3-month period of doing the history assignment. A survey instrument was used with a semistructured interview schedule. Broad interview questions were designed to elicit (a) how the student felt about his history assignment (e.g., “How do you feel about your history assignment at this point?”); (b) how they thought up the assignment (e.g., “How did you come up with your topic?”); and, finally, (c) how they obtained information (e.g., “How did you search for information on your topic? Did you ask anybody for help?”). In interview 6, the last interview, which took place the day after the student participants had presented the final version of their assignment, the students were also asked to make suggestions about the process of doing the assignment (e.g., “What type of help would you have liked?”).

The students were interviewed for approximately 10 minutes during class in their classroom on six separate occasions in the March–May 2011 period, most often on two separate days per interview point because the two classes were held on different days of the week. The assignment process from beginning to end lasted approximately 14 weeks. There was a 2-week spring break in the middle of March. The six interview points were: interview 1, March 1 and 3; interview 2, March 29 and 31; interview 3, April 14; interview 4, April 20 and 28; interview 5, May 4; and interview 6, May 27th.

Data Analysis

The transcripts of the interviews were analyzed utilizing Glaser and Strauss's (1967) grounded theory approach.

Glaser and Strauss's method is designed to lead to middle-range model/theory making in an exploratory or new area of study. Atlas.ti (v.7), a qualitative data analysis software program, was used to code and analyze the data. For the sake of measurement validity, the use of a software program facilitates, but does not ensure, objective counting of coding categories. For limitations of the coding scheme, particularly concerning validity and reliability, the reader is directed to the Limitations section of this article.

The intention of the data analysis was to explore the knowledge construction process of the 10 study participants as it evolved from beginning to end of the students' researching and putting together their history assignment. Because the students' answers to the interview questions were usually short and vague, the intention was to code every utterance by (a) ISP stage, (b) whether the utterance signified a feeling, thought, or action, and (c) also to attempt to dimensionalize the utterance in terms of implicit knowledge, which we had utilized previously in this research project (Cole et al., 2013). The dimensionalization coding took time to actualize in terms of knowledge construction; only late in the coding process did knowledge construction become the core coding category of the data analysis. Table 2 contains this initial raw data collection phase of coding, at the beginning of data analysis, when it still was not clear what the analysis would lead to.

In Glaser and Strauss's coding method, there are three types of coding, which denote phases of the coding process leading to model/theory making: step 1: open coding (when the researcher is exploring the data); step 2: axial coding (when the researcher identifies the "core" phenomenon); and step 3: selective coding (when the researcher selectively develops and codes propositions or hypotheses that link into the developing model).

Step 1: Open coding. The open coding framework at the beginning of coding was Kuhlthau's seven-stage ISP model. There are three levels or orders to this initial open coding:

- *First-order coding:* coding the seven stages of the ISP model: initiation, selection, exploration, focus formation, collection, presentation, and assessment
- *Second-order coding:* coding the student participants' mentions of feelings, thoughts, and actions over the seven stages of the students' ISP
- *Third-order coding:* coding the descriptive terms in Kuhlthau's ISP model to describe the students' feelings, thoughts, and actions; these descriptors are listed in Table 1 (above).

For second- and third-order coding, the students' interview answers could indicate a combination of coding categories and thus would be coded utilizing all appropriate terms (i.e., there was no attempt to select the "best" coding term only). For example, if the student participant indicated in the interview that he was now selecting their history topic, that the topic was vague in their minds, and that they felt

uncertain about it, but that they were optimistic they would eventually be able to make the selection, we coded the answer with the multiple coding terms: selection, feeling, thought, vague, uncertainty, optimistic. Again, the purpose of this initial phase of open coding was number crunching shown in Table 2—trying to see patterns in the data through a quantitative analysis of the answers the student participants gave in their interview.

Step 2: Axial coding around core category. As the coding of the transcripts of the interviews progressed, the emerging center of coding interest became the students' knowledge construction process over the course of the seven stages of their ISP. This became the axial core category for the study. We saw the coding process as going deeper and deeper, so we labeled the core category the fourth-order coding:

- Fourth-order coding: coding the student participants' mentions of the axial core coding category: knowledge construction and its dimensions over the seven stages of the students' ISP.

Step 3: Selective coding. Once the first-, second-, and third-order coding was finished for the 10 interview participants for each of the six interview points, the 10 interviews for each of the six interview points were then coded for the fourth-order axial core coding category (knowledge construction) and its dimensions (shown in Table 3).

Results

We structure the Results section according to the study's three research questions.

Research Question 1: Do the data support Kuhlthau's ISP model stages?

Research question 1 was a straightforward attempt to anchor the empirical findings in Kuhlthau's seven-stage ISP model. It duplicates the initial first-order open coding categories, which were based on the seven ISP model stages: initiation; selection; exploration; formulation; collection; presentation; and assessment. Highlighted (in bold) in Table 2 is the diagonal aspect of the results.

Table 2's first-order coding produces an evolving pattern that generally follows Kuhlthau's ISP model stages. The exceptions are stage 1 (initiation) and stage 5 (collection), where there was no discernible indication of the ISP model pattern in the data. Regarding the absence of data for stage 1 (initiation), in the third-order coding for the coding term "interest," there were three mentions by the study participants in interview 1 concerning their interest-feelings. We will refer to this at greater length in the Discussion section. Regarding stage 5 (collection), for the purposes of this study, collection was narrowly defined as the collection of evidentiary information in support of the students' thesis statement (a requirement for the participants' course assignment). Because there were so few mentions of this specific coding

TABLE 3. Progression over six interviews of Kuhlthau's seven-stage IPS model: second-order coding for feelings, thoughts, and actions, with selective third- and fourth-order coding.

Second order	Both feelings and thoughts			Both thoughts and actions		
	Feelings	Third-order uncertainty	Thoughts	Fourth-order knowledge construction	Third-order information seeking	Actions
Interviews(1–6)						
1. March 1/3	21	10	46	66	58	57
2. March 29/31	9	10	47	56	35	40
3. April 14	0	1	21	26	21	24
4. April 20/28	13	13	26	29	15	24
5. May 4	8	9	24	24	8	14
History presentation, May 26						
6. May 27	6	6	43	49	71	77

Arrows show up- or downward variance pattern from interview to interview.

category, in Table 2 in the collection column, we added a third-order code (information seeking), which we defined as all information seeking (i.e., a broader term than collection), to see whether it would support the diagonal aspect of the results. This was not the case. For interviews 4 and 5, when the ISP collection stage 5 should have logically occurred, information seeking was coded only 15 and eight times, respectively, compared to 58 times in interview 1. The participants appeared to have front-loaded their information seeking in the first two interview periods.

Research Question 2: Do the data support the evolution of the Kuhlthau ISP model's feelings, thoughts, and actions?

Research question 2 was, again, a straightforward attempt to anchor the present study in Kuhlthau's seven-stage ISP model. In effect, it duplicates the initial second-order coding categories which were based on the seven-stage ISP model feelings, thoughts, and actions. In order to further the analysis, Table 2 also includes the third-order coding category of uncertainty and information seeking and the fourth-order coding category of knowledge construction.

In Table 3, we have separated out interview 6 from interviews 1–5 because this last interview took place after the students had presented the final version of their history assignment to their instructors. It constitutes a separate assessment phase (see following for further discussion). During the stage 7 assessment interview 6 on May 27, for the second-order coding category (actions), the students were able to recount what actions they did in greater detail the day after the presentation of the assignment than during the actual time when they did the actions in interviews 1–5.

In Table 3, we indicate patterns in the results with arrows. Arrows pointing downward indicate a decrease in mentions by the study participants compared to the week(s) before; arrows pointing upward indicate increases. The arrows are designed to visually show patterns in the results. Table 3 results indicate that:

- (a) The participants' feelings were most pronounced in the first interview, then decreased in interviews 2 and 3; this was followed by an uptick in feelings in interview 4.

- (b) The participants' thoughts and actions occurred mostly in the first two stages of their ISP, then decreased in interviews 3–5.

- (c) For the third-order coding category of uncertainty, which includes both thoughts- and feelings-uncertainty, there was an uptick in participant mentions in interview 4, followed by a decrease in interview 5. Of the 13 mentions of uncertainty in interview 4, 11 are feelings-uncertainty and two were a combination of feelings- and thoughts-uncertainty. We will discuss this at greater length in the Discussion section.

- (d) For the fourth-order coding category of knowledge construction, which includes instances from both thoughts and actions, there were significant participant mentions in interviews 1 and 2, then a decrease in subsequent interviews.

- (e) For the third-order coding category of information seeking, which also includes instances from both thoughts and actions, there were significant participant mentions in interview 1 with a steady decrease in subsequent interviews.

Research Question 3: How can we chart the study participants' implicit knowledge-to-explicit knowledge construction, including the vital preparatory priming function, as it evolves over the ISP stages?

Table 4 lists the study findings for the fourth-order axial core code (knowledge construction), as well as for the dimensions of knowledge construction, specifically implicit-to-explicit knowledge construction. The assumption underlying research question 3 is that the knowledge construction process for the 10 study participants can be described by these dimensions as they evolve over the ISP stages.

From Table 4, knowledge construction is the broader or axial core coding term. The eight dimensions are narrower terms. The definitions of Table 4's eight dimensions (object is listed twice) are:

- *Object*: student mention of the assignment as a finished object
- *Implicit 1*: student mention of earlier interest in their assignment topic and/or another course taken previously concerning the assignment topic

TABLE 4. Fourth-order coding axial coding: knowledge construction (KC) and its dimensions over the six interviews.

	Knowledge construction (KC)									
Fourth order	Dimensions									
Interviews	KC	Object	Implicit 1	Priming	Implicit 2	Implicit 2 to 1	Aspects	Object	Jigsaw	Belief
1. March 1/3	66	30	18	16	40	5	7	30	0	0
2. March 29/31	56	21	6	8	39	14	11	21	6	1
3. April 14	26	9	1	4	12	11	6	9	2	0
4. April 20/28	29	15	1	0	15	11	7	15	3	1
5. May 4	24	12	1	0	12	6	0	12	7	8
History presentation, May 26										
6. May 27	49	30	4	9	28	13	2	30	14	4

- *Priming*: student mention of acquiring background information on topic area
- *Implicit 2*: student mention of knowledge construction procedures, such as referring to subtopics in the assignment, finding definitions of key topic concepts, drawing conclusions, making comparisons, and identifying cause-and-effect relations.
- *Aspects*: student mention of issues, problems, or controversies in their topic area.
- *Implicit 2 to 1*: student mention of applying automatic information processing procedures from implicit 2 to their previous knowledge of the topic from implicit 1 (previous interest in the topic or from a previous course in the construction of assignment)
- *Object*: student mention of the assignment as a finished object
- *Jigsaw*: student mention of formulating and/or putting together pieces of their topic into a structure that supports the assignment's focus thesis.
- *Belief*: student mention of constructing the assignment as part of their belief system

Table 4 orders the dimensions over the 3-month period of the study, from object, implicit 1, priming, and implicit 2, which are highly mentioned in the first two interviews, to aspects and implicit 2 to 1, which are highly mentioned in the middle interviews, and, finally, to object, jigsaw, and belief, which are highly mentioned in, respectively, interviews 4 and 5, interview 5, and interview 5. Interviews 4 and 5 were the last two interviews before the study participants' May 26 presentation of the final version of their history assignment.

Interview 6, which occurred on May 27, the day after the assignment presentation, corresponds to stage 7, the assessment stage of the ISP model. This was when the student could look back on the assignment process and reflect upon it. Compared to the previous five interviews, notable variances in participant mentions in assessment interview 6 were:

- Knowledge construction was heavily mentioned.
- Implicit 1 was lightly mentioned.
- Implicit 2 to 1 was heavily mentioned.
- Object was heavily mentioned.

- Jigsaw was very heavily mentioned, compared to the previous five interviews.

Discussion

Research Question 1: Do the data support Kuhlthau's ISP model stages?

From Table 2, except for ISP stage 1 (initiation) and stage 5 (collection), the results generally followed the stages of the ISP model. Stage 1 (initiation) may have been an extremely short stage for the study participants that occurred before our first interview point in the second week of the course.

Speculatively, coding around the term interesting during interview 1 when ISP stage 1 (initiation) should have been taking place, indicate potential for further research. We coded interesting seven times for interview 1. But, the coding divided between feelings and thoughts. In the first quote from the data, the feeling of interest as fun, cool, and enjoyable is indicated:

I think it is a fun project 'cause I want to learn certain topics that I actually get to choose instead of the teacher usually giving us a topic, so I find that interesting, and I like the fact that we can be in partners and discuss things and find a topic sentence (i.e., thesis statement). I find that pretty cool. (interview 1, student 10)

"Interesting" as a coding term also appears to have a cognitive aspect to it. For a student who was far ahead of the others in terms of constructing the history assignment, one of his/her advantages seemed to be a pre-existing interest in the topic of the assignment:

Yes, I knew a bit. For my tenth birthday my parents brought me to the Titanic exposition that came to [a Canadian city]. It fascinated me. I thought it was really interesting so I decided I wanted to learn even further. (interview 1, student 6)

Additionally, from Table 2, there appeared to be no separate stage 5 collection of information stage, which, in coding, we defined narrowly as collecting evidentiary information in support of the participants' thesis statement. When we included the broader code (information seeking), the

surprising and important result was the study participants' frontloading of their information seeking in interviews 1 and 2, which would correspond to ISP stages 1, 2, and, possibly, 3, but not to ISP stage 5 (collection). The concept of a thesis statement as a personal perspective, which was only one among other alternative perspectives on their topic and thus needed to be defended with the support of evidentiary information, did not appear to be ingrained in the minds of the 14-year-old Grade 8 students in our study.

Research Question 2: Do the data support the evolution of the Kuhlthau ISP model's feelings, thoughts, and actions?

Table 3 lists the second-order coding for feelings, thoughts, and actions of the study participants over the six interviews. For feelings, the greatest quantity of participant mentions occurred during interview 1, with a second uptick in mentions during interview 4. The interview 4 time period coincided with an intense intellectual period of their ISP when the students were trying to formulate their topic focus because they had to hand in a proposal containing their thesis statement a week later on May 4. The uptake in third-order coding for uncertainty for interview 4 therefore is predicted by the ISP model.

For thoughts and actions, study participants frontloaded their mentions in the first two interview points, with a steady decrease in mentions through interviews 3–5. For actions, this corroborates the ISP model as Table 2 indicates the participants were in stage 3 (exploration) in interview 2. However, for thoughts, the Table 3 results appear to contradict the ISP model, which predicts intense thinking in interviews 3–5 as students make the transition from exploring information in ISP stage 3 to focus-thesis formulation for their assignment in stage 4.

For the purpose of furthering the analysis, in Table 3, we also included the third-order coding category of information seeking and the fourth-order axial core coding category of knowledge construction. The results from these coding categories show front-ended intensity in interviews 1 and 2 with a trail-off in subsequent weeks. According to Table 2, stage 3 (exploration) occurred in interview 2 for the study participants, predicting intense information seeking in interview 2; Table 3 therefore corroborates this result. However, the Table 3 result for knowledge construction with front-loaded mentions in the first two interviews appears to contradict the ISP model according to Table 2, which puts ISP stage 4 (focus formulation) later in the interviews, in interview 4.

For the assessment ISP stage 7, the interview participants became much more introspective and able to articulate their thoughts and actions. From Table 3, this was particularly the case for the second-order coding category (actions) and the third-order coding category (information seeking), with 77 and 71 mentions, respectively, compared to the highest number of mentions in interview 1 with 57 and 58 mentions, respectively. It was as if the participants had been released from a burden in interview 6 (i.e., they had presented their assignment the day before) and now

they could allow themselves to talk about the assignment process. However, for the students' feelings, the opposite was the case. In interview 6, the study participants could not go back in time and remember their feelings compared to the overwhelming nature of their feelings as they were actually doing the assignment, especially early on in their ISP.

An interesting result from Table 3 concerns student uncertainty. In the conceptual framework section of this article, we emphasized Kuhlthau's uncertainty principle and its change in nature between feelings and thoughts depending on when study participants are asked to rate their uncertainty. We coded all references by the student participants to synonymous or semisynonymous terms of uncertainty as uncertainty (e.g., doubt, stress, and confusion). Among all the six interview points, interview 4 had the most mentions of uncertainty, which supports findings from previous ISP studies. Of the 13 codings of uncertainty in interview 4, 11 were feelings-uncertainty and 2 were both thoughts- and feelings-uncertainty. Here, we quote from one example of thoughts- and feelings-uncertainty:

Hum, I think, I mean, I think that what is causing me stress is finding a thesis and, hum, not finding a thesis, more research and choosing which person that I want, well, you know my thesis, so choosing which person I want to base my whole project on. (student 9, interview 4)

From the quote, this student participant's feelings-uncertainty centered on not finding a thesis for the assignment. On the other hand, the thoughts-uncertainty centers on choice and choosing. Having to choose something creates thoughts-uncertainty, and this student infers that this uncertainty will decrease when the choice has been made.

From the quoted instance of thoughts-uncertainty, there is very little evidence to draw conclusions about when in their ISP the Grade 8 student participants had thoughts- and/or feelings-uncertainty, which was a primary focus of this article's conceptual framework. However, for purposes of directing future research on this important topic, we will attempt to make several conjectures.

Polanyi (1966) states that subception and the functioning of proximal and distal terms serve to pick up new information about the world and then integrate this new knowledge about the world into our intellectual system:

Moreover, we keep expanding our body into the world, by assimilating to it sets of particulars which we integrate into reasonable entities. Thus we do form, intellectually and practically, an interpreted universe populated by entities, the particulars of which we have interiorized for the sake of comprehending meaning in the shape of coherent entities. (Polanyi, 1966, p. 29)

In the quote, Polanyi utilizes the terms "sets of particulars" and "coherent entities." Let's relate these two terms to his concepts of priming and subception, discussed in the conceptual framework section of this article. Priming and

subception allow the individual to acquire from the outside world the proximal term, which, according to Polanyi, is made up of multiple particulars, called the sets of particulars. The distal term, in turn, shocks the proximal term into a coherent entity. In the case of the Grade 8 student just quoted who indicates an example of thoughts-uncertainty, let us conjecture that the “entity” is this student’s thesis on his or her history topic. One set of the sets of particulars he or she mentions in the quote is “choosing which person I want to base my whole project on.” The student infers that he or she has a set of persons from which to choose, so we can conjecture that one of the sets of particulars is this set of persons in the student’s mind. There are other sets of particulars that make up this student’s thesis entity, but choosing the person from the set he or she has in mind is obviously an important part of this student’s thesis. We make the conjecture therefore that thoughts-uncertainty increasing has something to do with the creation in the student’s mind of a set for each of the particulars of the thesis entity, and that the thoughts-uncertainty is decreased when the student makes a selection from the set.

Research Question 3: How can we chart the study participants’ implicit-to-explicit knowledge construction, including the vital preparatory priming function, as it unfolds over the ISP stages?

Research question 3 investigates the dimensions of the process of knowledge construction, which is the axial core coding category for the study. Table 4 indicates the progression of the knowledge construction process of the study participants through eight dimensions, grouped into three phases. Here, we tentatively label the three phases in terms of the students (1) making, (2) asking, and (3) answering questions. We also illustrate each of the three phases with illustrative quotes from the student interviews.

Students Making Questions from their Previous Knowledge

After the student participants selected their assignment topic, there was an immediate concern for the final presentation of the assignment in 3 months (object), which influenced their thinking and actions. They also brainstormed by starting from what interested them (implicit 1), then automatically “making questions” based on their previous experience of life and how the world works (implicit 2). In this first phase, they also frontloaded their information seeking, exploring information on their topic through the Internet to gradually set up priming for the topic. There is a certain feeling of interest—scary, exciting, or overwhelming—because the exploration of information touches on implicit 1 (previous experience with the topic). Speculatively, we highlight this “feeling” aspect of interest.

To illustrate phase 1, we give an illustrative example. Here, the student engages in a type of automatic “making questions,” based on previous experience of making meaning from life experience, which the student then

applies to doing the assignment topic in order to begin to achieve meaning out of the topic. In other words, the student makes questions, which begins the creation of meaning and the process of knowledge construction for the history assignment:

I researched a bit about it and it led me to [the topic] “terrorist attack” and so maybe I will be thinking how terrorists are motivated to like killing and, like, why? There can be some mental issue with them but why? It could an innocent person so why would you kill that person for being innocent. (interview 1, student 9)

Topic Information Asking the Student Questions

At a certain point, however, topic information begins to raise questions about the topic in the student’s mind. The topic information becomes cognitively “interesting.” New information containing elements of knowledge construction is beginning to make an impact—that is, aspects of the topic itself infiltrate knowledge construction (aspects). Speculatively, we highlight the “cognitive” aspect of interest aroused in the student in this phase by the topic information asking the student questions.

Student answer: I was thinking of doing something about chess and its, hum, influence or connection with warfare.

Question: How did you come up with that?

Student answer: I don’t know. Playing chess since I was 5 years old and I’ve always loved playing chess. Thought I’d do something with that. . . . Well chess has to do a lot with thinking: If you make your move too quickly, you can lose your pieces. And the pieces also have their symbolic, for different classes of people. So yes, it could really have a lot to do with actual warfare. (interview 2, student 5)

Students Answering Questions the Newly Found Topic Information Asks the Student

In the third and final phase, pieces of a jigsaw puzzle have a structural quality (i.e., blue pieces signify a sky that can be grouped together, green pieces signify a lawn, reddish brown pieces signify bricks of a house, and so on). This realization that the aspects of the topic are actually pieces of a jigsaw puzzle centers on the realization of the puzzle theme, focus, or thesis. The student realizes: Ah, this puzzle is about a garden party on a great lawn in front of a 19th-century manor house. In other words, the thesis or focus structures the aspects of the jigsaw into a coherent object (object). And, the student can deepen the process of answering the questions asked by the newly found information on the topic:

Because it is an interesting topic and the more you research it the more you find out about, the more you find out about medicine’s past. You just start, hum, developing your thesis statement more and you start coming up with an answer to your thesis statement. (interview 5, student 4)

ISP Stages	1. Initiation	2. Selection	3. Exploration	4. Formulation	5. Collection	6. Presentation	7. Assessment
Feelings	Uncertainty	Optimism	Confusion/ Frustration/ Doubt	Clarity	Sense of direction/ confidence	Relief/ Satisfaction Or disappointment	Sense of accomplishment
Thoughts	General/ Vague			Narrowed/ clearer	Interest up	Clearer/focused	Increased self- awareness
KC	Implicit 1 PRIMING Implicit 2		Implicit 2 to 1 Aspects		Object JIGSAW Belief		Increased knowledge of belief system
	Making questions		Asking questions		Answering questions		
Actions	Seeking Background information		Seeking relevant information		Seeking relevant or focused information		

FIG. 4. Developing knowledge construction (KC) model inserted as a row in Kuhlthau's seven-stage ISP model.

Creating a coherent object—utilizing newly constructed knowledge as explicit knowledge to create a structured assignment centered on a thesis—out of the assignment topic, creates an inner feeling of coherence as well, with feelings and cognition flowing together. The student is actualizing his belief system (“This is what I truly believe about this topic.”). The students link what they were doing in their assignment to their belief system about how the world works by answering questions raised by their assignment topic:

Because I learn a lot about it and, hum, I can relate to it because it is about child labor and I am a child like I can see what it feels, well, kind of imagine what it feels like. (interview 5, student 10)

We emphasize that this process of answering questions not only actualizes their belief system, but also develops their belief system. Speculatively, we highlight the affective and cognitive aspects based on interest of the student as the student feels an intense sense of accomplishment and heightened self-awareness because of the actualization of their belief system resulting from the knowledge construction process.

An Emerging Knowledge Construction Model

Kuhlthau's ISP model describes the role played by students' feelings and thoughts in motivating their information behavior actions as these feelings, thoughts, and actions evolve over the six stages of completing the students' school assignment, and the seventh stage, which is an assessment stage after assignment completion. Kuhlthau's assessment seventh stage defines students' feelings as a sense of accomplishment and the students' thoughts as increased self-awareness. In effect, these stage 7 definitions define the end game of knowledge construction in the ISP model. The definitions therefore signify a uniquely personal affective-

cognitive evaluation measure of the students' learning as a result of seeking information for their class assignment. It is not an objective measure.

Following Kuhlthau's lead in her assessment stage 7's end game definitions for the students' feelings and thoughts, we define the knowledge construction's end game as:

- The addition to or modification of the students' belief system as a result of completing the class assignment and
- The students' actualization of their belief system through the new topic information they seek, find, and use over the course of completing the assignment. In other words, the students know better not only what they believe to be true about the world, but also a sense of coherence about where they stand in relation to the world.

In Figure 4, we tentatively insert knowledge construction as a separate row in Kuhlthau's seven-stage ISP model. However, for this row, instead of ISP stages 1–6, we retain the three stages of knowledge construction outlined in the previous Discussion section. In ISP assessment stage 7 in this added row, we define the end game of knowledge construction as increased student knowledge of their own belief system.

Limitations

This was a qualitative, exploratory study. Its purpose was to create a coding scheme for future studies utilizing the same data collection instrument. Reliability, particularly intercoder agreement, is of utmost importance in these future studies. Concerning external validity, the 10 study participants in the present study were a self-selected sample of Grade 8 students in a private school where tuition fees may limit access. The optimal nature of the study participants' school and parental learning environments compared to the larger population of North American 14 year olds

increases the percentage of study participants who have achieved the Piaget formal operational stage of cognitive development. The results of this study therefore cannot be considered statistically representative of any larger population.

Conclusion

The present article reports on a qualitative study of 10 Grade 8 students researching a real-life history assignment. The students were interviewed at six different points over the 12-week duration of the assignment, from a week after the initiation date when the instructor announced the assignment to the day after the students presented the final version of the assignment for marking. The conceptual framework for the study is Kuhlthau's seven-stage ISP model, which includes an assessment stage when the students look back over the assignment construction process.

The results of the study generally support previous ISP studies, except for two results, which we highlight here:

1. From Table 2, the study participants front-loaded their knowledge construction activities in the first 4 weeks of the 3-month assignment process, instead of in ISP stage 4 (focus/thesis formulation), which the Table 2 results indicate occurred for the participants in the middle weeks of the 12-week assignment. This result also appears to contradict previous ISP studies.
2. From Table 2, the study participants did not appear to have a stage 5 (collection stage), which we defined narrowly as collecting evidentiary information to support their focus-thesis statement. Even if broadly defined, as information seeking, the study participants frontloaded information seeking actions in the first 4 weeks of the 3-month assignment process, not in the middle and late stages as predicted by the ISP model.

The results also indicate new directions for ISP studies, specifically the results concerning the evolution of knowledge construction of the study participants over the seven stages of their ISP. These new direction results provide a three-phase model of the knowledge construction process:

1. The students make questions from their previous knowledge of life and how the world works in order to start knowledge construction.
2. The topic information the students have found begins to ask the students questions. Uncertainty may rise as a result of the formation of a question set. This could be a set of possible historical persons they could analyze in some way as part of their topic.
3. In the third and final knowledge construction phase, the students select the questions they want to answer. The students answer questions, deepening their thesis into their belief system.

We have attempted to operationalize the three-phase knowledge construction model, in terms of making, asking, and answering questions, as an added row to Kuhlthau's ISP

model. Operationally defining knowledge construction over the course of the ISP model stages, in terms of the evolution of the questions, has the potential of facilitating the development of information systems for this particular user group and task.

More speculatively, the study findings indicate future directions for further research based on Kuhlthau's uncertainty principle and the feeling-thought duality of this important information science concept. The findings speculatively suggest that the students' uncertainty switches from the feeling of uncertainty to a cognitive form of uncertainty in the middle phase of the three-phase knowledge construction process. This switch may be caused by the formation of the thesis-question set that occurs in this middle stage of knowledge construction. The thesis-question set forms when the students transition from making questions from their previous knowledge to the information the students have newly found on their topic asking the students questions. In our conjecturing in the Discussion section, based on a discussion of Polanyi (1966) and from a student quote illustrating thoughts-uncertainty, it is possible that the selection of the thesis question is a function of making selections from sets of particulars, one set of which, from the student quote in the Discussion section, is the set of alternative historical persons the student is considering upon which to base his or her project.

The study findings also speculatively suggest that the students' feelings of interest in their topic becomes cognitive interest at a certain point. In the final phase of the knowledge construction process, students may link their knowledge construction to their belief system. At this point, the students' feelings of interest and cognitive interest flow together, creating a sense of accomplishment and self-awareness, which constitutes the end game criteria for Kuhlthau's assessment stage seven's feelings and thoughts, respectively.

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